



DEPARTMENT OF
COMPUTER SCIENCE



PSEUDOCODES I

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LEARNING OBJECTIVES

At the end of the lesson, you should be able to:

- Explain what pseudocodes are and some of the rules that govern them.
- Identify various types of pseudocode constructs.
- Discuss the advantages and disadvantages of pseudocode.



PSEUDOCODE

A pseudocode is a textual representation of the sequence of operations in a computer program; or the textual representation of an algorithm.

Pseudocodes are written in a form that makes them easily convertible into programming statements.

The main motivation behind the invention of pseudocode is to be able to write all the main steps of an algorithm in a language that is very similar to English language.



PSEUDOCODE

A rule of thumb is that if you want to develop a new computer program and you are struggling with how to start the development, write a pseudocode first.

- A common technique is to firstly, write pseudocode like comments in a file.
- After writing the pseudocodes, we can now start to fill in between the comments with the actual code.
- By following this approach, we are essentially writing the comments for our program before writing the program.



PSEUDOCODE RULES

Each line in a pseudocode should only contain one statement. Since a pseudocode deals with giving the computer a list of tasks to implement in a sequential manner, it is better to ensure that each task is written on a separate line.

Write keywords in capital letter. In order to differentiate keywords from other words in the pseudocode, we can also ensure that keywords are written in capital letter.

Pseudocode statements should be language independent; not in a particular programming



PSEUDOCODE CONSTRUCTS

Pseudocode constructs (also known as pseudocode keywords) are used to outline the control flow of an algorithm. Some of the constructs are;

- SEQUENCE
- CASE
- WHILE
- REPEAT-UNTIL
- EXCEPTION and WHEN
- FOR
- IF-THEN-ELSE
- CALL



SEQUENCE

The SEQUENCE keyword represents tasks that are linear and performed sequentially one after the other. We will see an example.

Input: READ, OBTAIN, GET

Output: PRINT, DISPLAY, SHOW

Compute: COMPUTE,
CALCULATE, DETERMINE

Initialize: SET, INIT

Add: INCREMENT, BUMP

Sub: DECREMENT



FOR

With the FOR keyword, one or more statements are executed up to a desired number of times, based on setting all the necessary conditions at the beginning/top.

```
FOR iteration bounds  
sequence  
ENDFOR
```

Let us see an example



WHILE

With the WHILE keyword, one or more statements are executed up to a desired number of times, based on setting some condition at the beginning/top and other conditions in the body of the statements.

```
WHILE condition  
sequence  
ENDWHILE
```

Let us see an example...



REPEAT-UNTIL

With the REPEAT-UNTIL keyword, one or more statements are executed up to a desired number of times, based on setting some condition at the ending/bottom and other conditions in the body of the statements.

```
REPEAT  
sequence  
ENDFOR condition
```

Let us see an example...



IF-THEN-ELSE

The IF-THEN-ELSE keyword uses conditional statements to change the normal flow of operation of an algorithm.

```
IF condition THEN  
    sequence 1  
ELSE  
    sequence 2  
ENDIF
```

Let us see an example



CASE

The CASE keyword also uses conditional statements to change the normal flow of operation of an algorithm. However, it is represented in a more generalised form.

```
CASE expression OF  
condition 1: sequence 1  
condition 2: sequence 2  
...  
condition n: sequence n  
OTHERS:  
default sequence  
ENDCASE
```

Let us see an example



CALL

The CALL keyword can be used to perform operations such as defining class objects and calling functions.

The subject of classes and functions will be discussed in future lectures.

CALL AvgAge with studentAges

CALL Swap with CurrentItem and TargetItem

CALL getBalance **RETURNING** aBalance

CALL SquareRoot with orbitHeight **RETURNING**
nominalOrbit



EXCEPTION AND WHEN

The EXCEPTION and WHEN keywords are used to handle exceptions.

When unexpected or unwanted events takes place when a computer program is running, responding to these events is called exception handling.

BEGIN

statements

EXCEPTION

WHEN exception

statements to handle the exception

WHEN another exception

statements to handle the exception

END



MID LESSON QUESTIONS

1. A pseudocode is a graphical representation of the sequence of operations in a computer program.
 - a) Yes
 - b) No



PSEUDOCODE ADVANTAGES

There is no need to use a special software to write a pseudocode.

It can be written with any text editor.

Anyone can understand pseudocode since the language is close to English language.

We can easily modify pseudocodes.

Structured concepts can easily be implemented with a pseudocode.



PSEUDOCODE DISADVANTAGES

Since there are no generally accepted standards for writing pseudocodes, anyone can write pseudocode in their own way, leading to various pseudocode styles and possible complications; especially if the pseudocode is not well written.

Compared to flowcharts that are visual and easily understandable, pseudocodes are not visual.

Also, compared to flowcharts, it will take a longer time for anyone to read and understand pseudocodes.



REFERENCES/FURTHER READING RESOURCES

- Goodrich, M. T., & Tamassia, R. (2015). Algorithm design and applications (Vol. 363). Hoboken: Wiley.
- Shaffer, C. A. (1997). A practical introduction to data structures and algorithm analysis. Upper Saddle River, NJ: Prentice Hall.
- Pseudocode: What It Is and How to Write It.
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THANK
YOU